



Chapter 14 — Budget, Capability & Historical Benchmarking Module



Feature summary: DOIP analyzes a defence organization's **budget and resource inventory**, runs scenario batches to find **capability gaps**, recommends **what to acquire within the available budget** (with evidence), and benchmarks every simulation against **documented real-world events from other countries** so planners can see “this is what happened elsewhere — here's how to prepare better.” Working name: **Preparedness Investment Advisor (PIA)**.

14.1 Concept & Motivation

Every run of the simulation engine already produces objective outcome metrics: response times, coverage, shortages, mission failures (Ch. 11 §11.6). This module closes the loop: it connects those outcomes to **money and inventory**, answering the questions decision-makers actually ask:

1. *Where does our budget currently go, and what does our inventory actually cover?*
2. *Which capability gaps caused the worst simulated outcomes?*
3. *Given a fixed budget, what should we acquire next for the biggest improvement in lives saved and response time?*
4. *When a similar situation happened in another country, what were the real outcomes — and are we on track to do better?*

The end goal is exactly as stated: **help defence organizations prepare better and save lives** — by making preparedness investment measurable, comparable, and evidence-based.

14.2 Honest Framing & Data Boundary



This module works exclusively with **public, open-source data** (§14.7) and **user-supplied inventories** (e.g., a training organization's own resource list). It never uses classified information, never models weapons effectiveness or targeting, and its objective functions are strictly humanitarian: response time, coverage, shortage avoidance, and a lives-saved proxy. Outputs carry the standard “AI/simulation-derived — decision-support only” label. This keeps the module within the platform's non-goals (Ch. 1 §1.8) while remaining genuinely useful for military preparedness planning and training. **Why the benchmark corpus is HADR-centric:** combat outcome data is classified in every country — military disaster-relief deployments are the only class of military operations with rich, public, legally usable outcome records, which is exactly what evidence-based benchmarking requires.

14.3 The Four Components

A. Budget & Inventory Analyzer

- Import budget and resource data via CSV/JSON templates (categories: personnel, vehicles, medical, communications, logistics, training, maintenance, reserves).
- Dashboards: allocation breakdown, year-over-year trends, cost-per-capability views, inventory coverage vs operational area (e.g., ambulances per sector, radios per team).
- Normalization layer maps imported line items onto DOIP's resource taxonomy so simulations can consume them directly — **the imported inventory becomes the starting state of scenarios**.

B. Capability Gap Analysis (simulation-driven)

- Run a **scenario battery** (10–30 seeded scenarios spanning the Ch. 19 military library: border surveillance, convoy escort, casualty evacuation, logistics sustainment, and HADR deployment) against the current inventory.
- For every degraded outcome, an attribution pass identifies the binding constraint: *“Response to Incident #12 exceeded target by 22 min → nearest*

unit lacked water transport", "Field hospital saturated → medkit stock exhausted at t+3h".

- Output: a ranked **Capability Gap Register** — each gap linked to the exact runs, events, and metrics that revealed it.

C. Recommendation Engine (budget-constrained optimization)

- Formulated as a **portfolio optimization / knapsack problem** (PuLP or OR-Tools CBC solver):
 - **Decision variables:** candidate acquisitions from an equipment catalog (each with cost, capability effects, quantity bounds).
 - **Objective:** maximize weighted improvement in simulated outcomes (lives-saved proxy, p95 response time, shortage hours avoided).
 - **Constraints:** total budget, category floors/ceilings, maintenance overhead (each item adds recurring cost), personnel/training prerequisites.
- Every candidate portfolio is **validated by re-running the scenario battery** with the upgraded inventory — recommendations are backed by measured before/after deltas, not assumptions.
- Output: 2–3 ranked portfolios (e.g., *balanced*, *medical-focused*, *mobility-focused*), each showing cost breakdown, projected outcome improvements with uncertainty bands, and links to the evidence runs.

D. Historical Benchmark Library ("what happened in other countries")

- A curated corpus of **documented real-world events** encoded two ways:
 1. **Case-study documents** (ingested into the RAG corpus): timeline, resources deployed, outcomes, official lessons-learned reports.
 2. **Analog scenarios** (YAML): a faithful DOIP scenario approximating the event's scale, geography, and resource situation.
- Starter case studies (all from public post-event reports): 2004 Indian Ocean tsunami response · 2010 Haiti earthquake logistics · 2015 Nepal earthquake airlift operations · 2018 Kerala floods · 2023 Türkiye–Syria earthquake USAR operations · 2011 Tōhoku evacuation. Each case is encoded around its **military response component** — airlift operations, army engineering, naval

evacuation, field hospitals, convoy logistics — the parts directly transferable to defence preparedness.

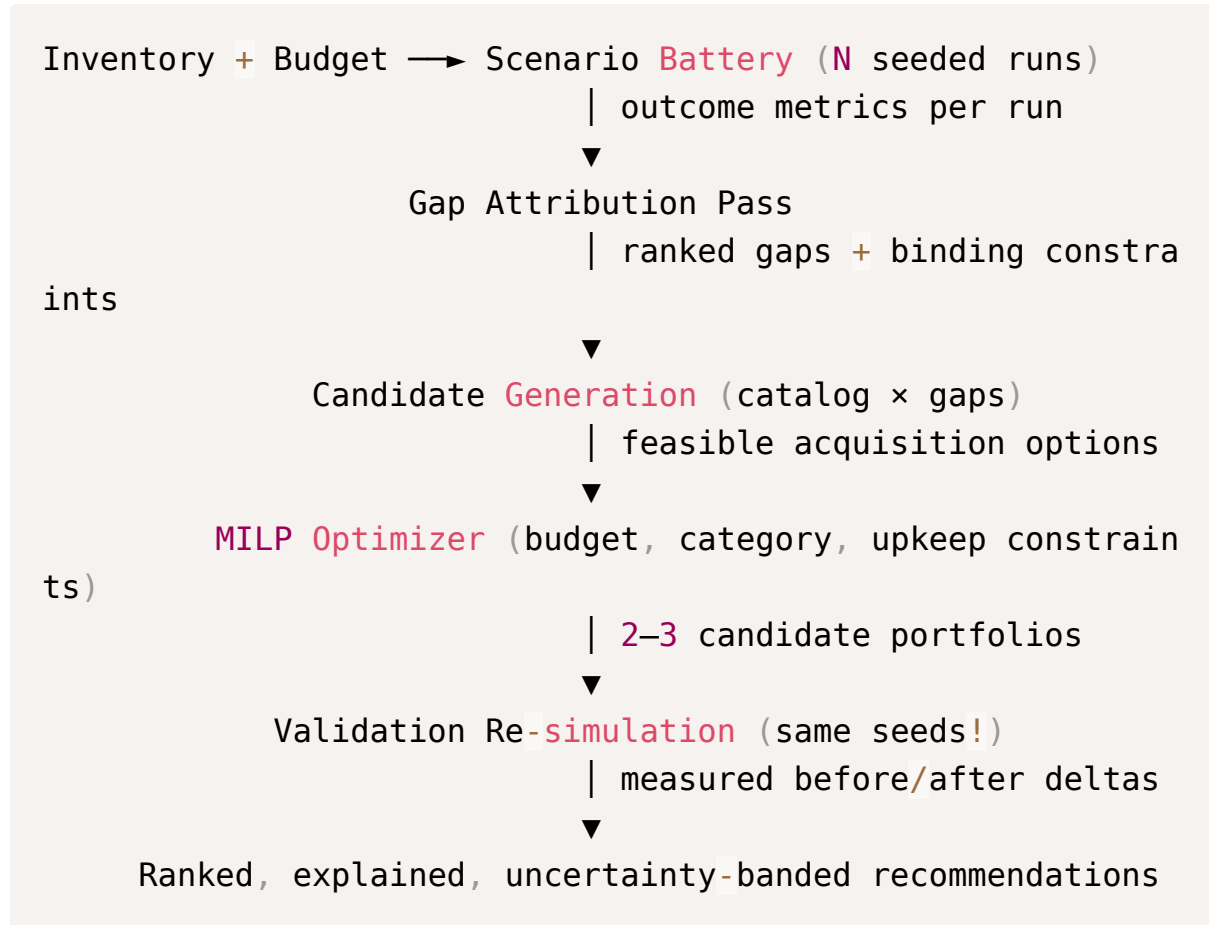
- **During and after any simulation**, a “Historical Parallels” panel matches the running scenario to similar cases (hazard type, scale, terrain, resource profile) and shows: what actually happened, real response timelines, documented failure modes, and cited lessons — side by side with the user's simulated metrics.
- **Calibration use:** running an analog scenario and comparing simulated vs documented real outcomes gives an honest error bar on the engine's realism — published in the after-action report.

14.4 Functional Requirements

- FR-14.1: Budget/inventory import via versioned CSV/JSON templates with validation and unit normalization; imports are auditable and reversible.
- FR-14.2: Allocation and inventory dashboards with category, trend, and geographic-coverage views.
- FR-14.3: One-click **scenario battery** execution (headless fast runs, Ch. 13 infrastructure) producing the Capability Gap Register with full run-level evidence links.
- FR-14.4: Equipment catalog CRUD (admin): items with cost, capability effect parameters, recurring cost, prerequisites.
- FR-14.5: Optimizer produces ≥ 2 candidate portfolios within budget; each is auto-validated by re-simulation; results show before/after metric deltas with uncertainty.
- FR-14.6: Every recommendation is **explainable**: click-through from “buy 4 water-transport vehicles” to the gap, the runs, and the historical cases supporting it.
- FR-14.7: Historical Parallels panel: similarity-matched case studies with cited excerpts (RAG with mandatory citations) shown during simulation and in after-action reports.
- FR-14.8: Benchmark mode: run an analog scenario and render simulated vs documented real outcomes on one chart, with divergence annotations.
- FR-14.9: All outputs labeled “Simulation-derived decision support — not procurement authority”; export to PDF/report for offline review.

- FR-14.10: Role gating: import and catalog management = admin; analysis and recommendations = planner and above; viewing = all roles.

14.5 Recommendation Engine — Design Notes



Key honesty mechanisms:

- **Same-seed validation:** before/after comparisons reuse identical scenario seeds, so deltas are attributable to the inventory change alone (determinism, Ch. 7).
- **Uncertainty:** each delta is reported across the whole battery (median + spread), never from a single run.
- **Lives-saved proxy:** defined transparently (incidents resolved within golden-hour targets, weighted by severity) — documented in-app, adjustable by the user, never presented as a real-world casualty prediction.
- **Sensitivity analysis:** ±20% budget sweep shows which recommendations are robust vs marginal.

14.6 “Similar Results From Other Countries” — Matching Design

- Each case study is fingerprinted on structured features: hazard type, affected population scale, terrain class, response-resource profile, season/weather.
- Scenario ↔ case similarity = weighted feature match + embedding similarity of narrative descriptions (reuses pgvector, Ch. 5).
- The panel always distinguishes **“documented fact”** (cited from the case corpus) from **“simulated projection”** (DOIP output) — different visual styling, no blending of the two.

14.7 Data Sources (all public/open)

Source	Used for
SIPRI Military Expenditure Database	Budget context and country comparisons
National budget publications (e.g., public defence budget documents)	Allocation categories and trends
EM-DAT (CRED international disaster database)	Event scale and impact statistics
UN OCHA / ReliefWeb situation reports	Response timelines and resource deployments
Post-event lessons-learned reports (govt/NGO, public)	Case-study corpus for RAG
World Bank Open Data	Population, infrastructure context
User-supplied CSV/JSON	The organization's own (non-classified) inventory

14.8 Acceptance Criteria

1. Importing the sample inventory + running the battery produces a Gap Register where every gap links to ≥1 concrete run/event as evidence.
2. The optimizer returns portfolios that never exceed budget and always pass constraint checks; validation re-simulation shows a measurable improvement on ≥70% of battery scenarios for the top portfolio.

3. The Historical Parallels panel surfaces ≥ 1 relevant case study for each starter scenario type, with working citations to corpus documents.
4. Benchmark mode renders simulated-vs-documented comparison for at least 2 analog scenarios, with divergences annotated rather than hidden.
5. Every recommendation screen carries the decision-support disclaimer and full evidence click-through.

14.9 Roadmap Placement & Effort

Slot	Work	Depends on
Phase 6.5 (after analytics) or Phase 8 (post-v1)	Sprint A: import + dashboards + equipment catalog · Sprint B: scenario battery + gap attribution · Sprint C: optimizer + validation loop · Sprint D: case-study corpus + parallels panel + benchmark mode	Engine & determinism (Ph. 1), headless runs (Ch. 13), logistics model (Ph. 5), analytics + after-action reports (Ph. 6), RAG (Ph. 3)

Estimated effort: **4 sprints (~6–8 weeks)** — the largest single module after the core platform, so it is deliberately scheduled after v1 is demonstrable. Authoring good case studies is the hidden cost: budget ~1 day per case for research and encoding.

14.10 Risks Specific to This Module

Risk	Impact	Mitigation
Overclaiming realism — users treat simulated deltas as real-world guarantees	High	Benchmark mode publishes the sim-vs-reality error bar; disclaimers on every output; livesaved proxy documented as a proxy
Garbage-in on imported budget/inventory data	High	Strict import validation, unit normalization, anomaly flags (e.g., impossible quantities)
Case-study bias (few cases → misleading parallels)	Medium	Minimum-similarity threshold before showing a parallel; “limited evidence” badge; grow corpus over time
Optimizer gaming the proxy metric	Medium	Multi-objective scoring; human-adjustable weights; portfolios

Risk	Impact	Mitigation
		always shown with <i>all</i> metrics, not just the objective
Political/organizational sensitivity of recommendations	Medium	Public-data-only boundary; educational framing; export reports clearly watermarked as simulation-derived
Scope creep into real procurement tooling	Medium	Non-goals restated: no vendor integration, no pricing negotiation, no classified data ingestion